Technical Architecture

1. Start

The system begins when a user accesses the DarshanEase web platform through a browser. This initiates the client-side application.

---

2. React JS

The frontend of the application is built using **React**, which enables a modular, component-based architecture for building interactive user interfaces. React handles dynamic rendering of temple listings, booking interfaces, and dashboards.

---

3. Client Application (React/Vite App)

The client-side application is powered by **Vite**, which provides fast development builds and optimized production bundling. This layer handles:

- UI rendering
- Routing between pages
- User interactions
- Sending API requests to the backend

---

4. User Interaction

This stage represents actions performed by the user on the interface, such as:

- Browsing temples
- Logging in or registering
- Booking darshan slots
- Viewing booking history
- Admin management operations

These interactions trigger API calls to the backend.

---

5. User Role Decision

The system determines the role of the user:

- **Public User (Unauthenticated)** – visitors who are browsing without logging in
- **Authenticated User (Devotee)** – users who are logged in and can book darshan slots
- **Admin User** – administrators who manage temples and slot schedules

Role-based routing ensures that users access only the functionalities permitted for their role.

Public Actions (No Authentication)

Public users can access limited features of the system:

- Browse temples
- View temple details
- Explore platform information
- Register or login to access booking features

These requests are sent to the backend API.

User Actions (Authenticated Devotees)

After authentication, devotees can perform booking-related operations:

- Browse temples
- Book available darshan slots
- View previously booked tickets
- Cancel bookings
- Manage personal dashboard

All these operations interact with the backend API for data processing.

Admin Actions

Administrators have elevated privileges that allow management of temple data and booking slots.

Admin functionalities include:

- Adding new temples
- Editing temple information
- Deleting temples
- Managing darshan slot availability
- Managing ticket capacity

Admin requests are also processed through the backend API.

API Layer (Express / Node)

The backend server is built using **Node.js** with **Express.js**.

This layer acts as the **central API gateway** for all system requests.

Responsibilities include:

- Handling HTTP requests from the client
- Implementing authentication logic
- Processing booking requests
- Managing temple and slot data
- Communicating with the database

Database Operations

The API performs CRUD operations on the database, such as:

- Creating bookings
- Retrieving temple data
- Updating slot availability
- Deleting or modifying records

These operations ensure data consistency and system functionality.

Database (MongoDB)

The application uses **MongoDB** as the primary database.

The database stores collections such as:

- Users
- Temples
- Slots
- Bookings

MongoDB's document-based structure allows flexible storage of temple information and booking relationships.

Data Response

After processing the request, the database sends the required data back to the API layer.

Client Response

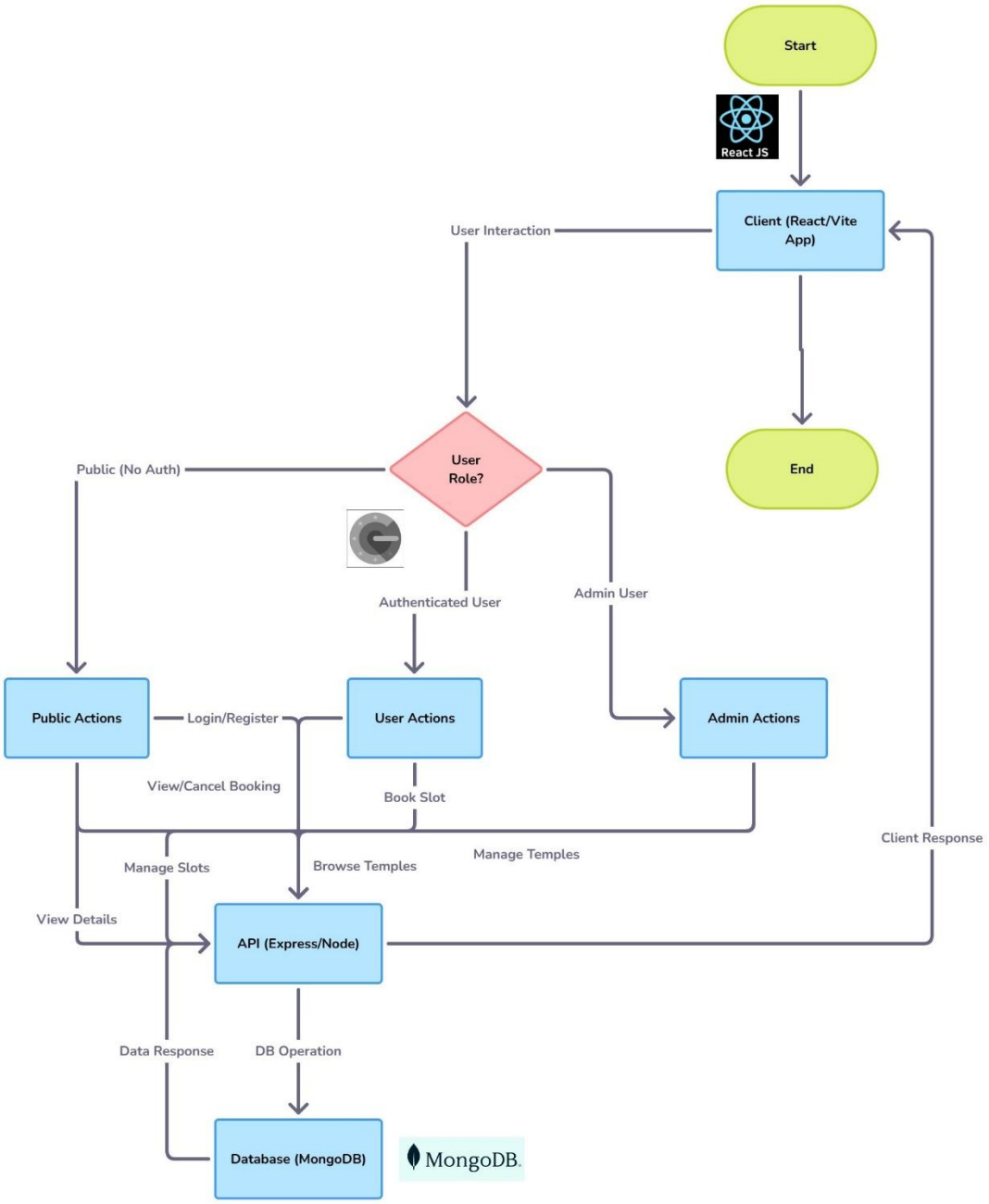
The backend API returns the processed response to the client application, where the React interface updates the UI dynamically.

Examples include:

- Booking confirmation messages
- Updated slot availability
- Booking history updates

End

The process completes when the updated data is displayed to the user on the frontend interface.


Start
React JS
Client (React/Vite App)
User Interaction
End
User Role?
Public (No Auth)
Authenticated User
Admin User
Public Actions
Login/Register
User Actions
Admin Actions
View/Cancel Booking
Book Slot
Manage Slots
Browse Temples
Manage Temples
Client Response
View Details
API (Express/Node)
Data Response
DB Operation
Database (MongoDB)
MongoDB.

ER Diagram

Entities in the System

The system consists of five primary entities:

1. **Regular User**
2. **Admin**
3. **Booking / View Bookings**
4. **Temples**
5. **Slots**

Each entity represents a major component of the temple darshan booking platform.

---

1. Regular User → Booking Relationship Relationship: has / does

Cardinality:
1 : N (One-to-Many) Explanation

- A **Regular User** can create **multiple bookings**.
- Each **booking record belongs to exactly one user**.

Example

A devotee may book:

- Morning darshan at Temple A
- Evening darshan at Temple B

Thus one user may have multiple booking records.

Attributes of Regular User

- user_id (Primary Key)
- email
- password

Attributes of Booking

- booking_id (Primary Key)
- user_id (Foreign Key)
- temple_id
- slot_id
- admin_id (reference for management)

---

2. Booking → Temple Relationship Relationship: of

Cardinality:
N : 1 (Many-to-One)

Explanation

- Many bookings can belong to **one temple**.
- A temple may have **many devotees booking darshan slots**.

Example

If 100 devotees book darshan at Tirupati Temple, then:

- Temple → 1
- Bookings → 100

Temple Attributes

- temple_id (Primary Key)
- name
- location
- description
- image

---

3. Temple → Slot Relationship Relationship: has / update

Cardinality:
1 : N (One-to-Many)

Explanation

- Each **temple has multiple darshan slots**.
- A slot belongs to **only one temple**.

Example Temple:

Kedarnath Slots

may be:

- 6:00 AM
- 8:00 AM

10:00 AM

- 12:00 PM

Slot Attributes

- slot_id (Primary Key)
- temple_id (Foreign Key)
- time
- capacity

---

4. Admin → Temple Relationship

Relationship: manages

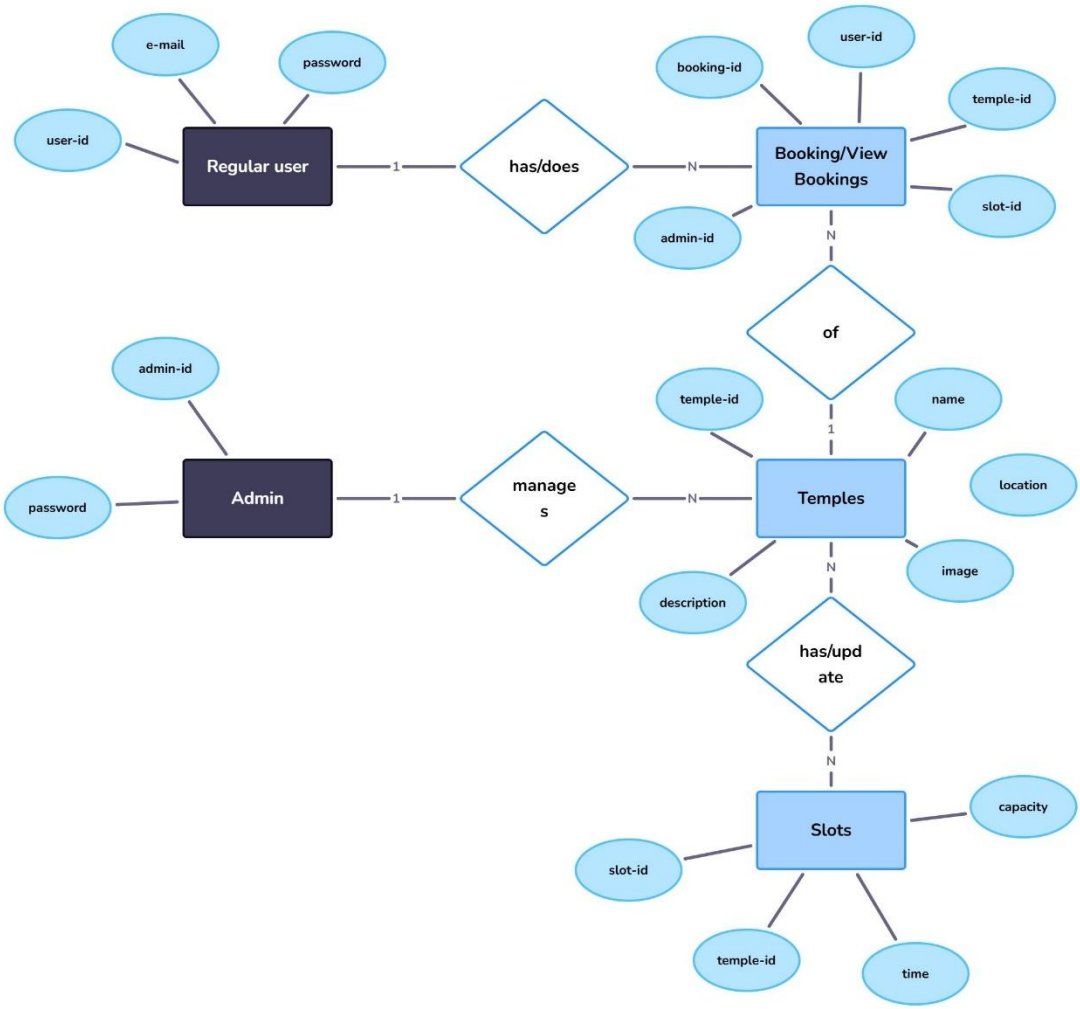
Cardinality:
1 : N (One-to-Many) Explanation

- An **Admin can manage multiple temples**.
- Management operations include:
  - Adding temples
  - Editing temple information
  - Deleting temples
  - Managing slots

Admin Attributes

- admin_id (Primary Key)
- password

•


e-mail
password
user-id
Regular user
1
has/does
N
booking-id
user-id
temple-id
Booking/View Bookings
slot-id
admin-id
N
of
1
admin-id
temple-id
name
Admin
password
1
manage s
N
Temples
location
image
description
N
has/upd ate
N
Slots
capacity
slot-id
temple-id
time

FEATURES

1. Devotee Registration and Authentication

The system provides a secure authentication mechanism that allows devotees to register and log into the platform. Users must authenticate themselves before accessing booking features. The authentication system verifies user credentials and maintains session security to ensure only authorized users can perform booking operations.

Key capabilities include:

- Devotee registration with email and password
- Secure login and logout functionality
- Authentication verification before booking darshan slots
- Role-based access control for devotees and administrators

---

2. Temple and Darshan Listings

The platform displays a comprehensive list of temples and available darshan slots. Devotees can browse temples along with details such as location, description, and available darshan schedules.

Displayed information includes:

- Temple name
- Temple location
- Temple description
- Available darshan dates
- Available darshan timings

This allows devotees to easily explore and select temples for booking.

---

3. Temple Selection

Users can select temples offering darshan slots and view detailed information about each temple. This feature helps devotees understand the temple's offerings and available darshan opportunities before making a booking.

Temple information displayed includes:

- Temple name
- Location
- Description

  Temple image
- Available darshan slots

-

---

4. Darshan Slot Selection

After selecting a temple, devotees can choose their preferred darshan slot from available time slots. The system displays all available slots and their capacities.

Features include:

- Viewing available darshan timings
- Slot capacity management
- Real-time slot availability updates
- Selection of preferred slot

---

5. Ticket Booking System

Devotees can book darshan tickets for their selected slot. The system records the booking and generates a unique booking ID.

Booking features include:

- Selection of darshan slot
- Booking confirmation
- Recording booking details in the database
- Support for multiple devotees per booking (if applicable)

---

6. Booking Confirmation

After successful booking, the system provides confirmation of the booking. This confirmation includes important information regarding the darshan visit.

Confirmation details include:

- Booking ID
- Temple name
- Darshan date
- Darshan timing
- Booking status

The confirmation may be displayed on the user dashboard and optionally sent via email.

---

7. Booking History and Management

Devotees can access a personal dashboard where they can view all their past and upcoming bookings. This helps users keep track of their darshan visits.

Features include:

- Viewing previous bookings
- Viewing upcoming bookings
- Canceling bookings (if allowed)
- Booking status display

---

8. Real-Time Darshan Slot Availability Tracking

The system continuously monitors the availability of darshan slots. Whenever a booking is made or canceled, the system updates the available slot capacity in real time.

Benefits include:

- Prevents overbooking
- Provides accurate slot availability
- Ensures fair allocation of darshan slots

---

9. Organizer Dashboard

Temple organizers are provided with a dedicated interface to manage temple and slot-related operations.

Organizer capabilities include:

- Managing temple information
- Managing darshan slots
- Viewing devotees and booking data
- Managing slot availability

---

10. Darshan Slot Management

Organizers are responsible for creating and managing darshan slots for temples.

Slot management features include:

- Creating new slots
- Updating slot timings
- Managing slot capacity
- Deleting slots when required

This ensures smooth operation of temple darshan scheduling.

•

---

11. Admin Dashboard

The system includes an administrative dashboard that provides administrators with complete control over the platform.

Admin functionalities include:

- Managing users and devotees
- Managing temple information
- Managing organizers
- Monitoring bookings
- Managing darshan slot availability

The admin dashboard ensures efficient management of the entire booking system.

---

12. Reporting and Analytics

The system can generate reports and analytics related to temple activities and bookings.

These analytics may include:

- Number of bookings per temple
- Popular darshan slots
- Devotee statistics
- Temple performance insights

These reports help administrators understand system usage and improve service management.

---

13. Integration with External APIs

The system can integrate with external APIs to enhance functionality and improve data synchronization.

Possible integrations include:

- Temple information services
- Real-time darshan slot updates
- Notification services
- Payment gateway integration for donations

---

14. Donation Integration (Optional Feature)

The platform can include an optional donation feature that allows devotees to contribute donations while booking darshan tickets.

Features include:

- Secure online payment integration
- Optional donation during ticket booking
- Recording donation transactions

---

15. Responsive and Interactive User Interface

The system provides a responsive and modern user interface designed to work seamlessly across desktops, tablets, and mobile devices.

UI features include:

- Responsive layout
- Interactive components
- Smooth navigation
- Real-time notifications

Libraries used include React, Bootstrap, and React Toastify.

ROLES AND RESPONSIBILITIES

1. Devotee (User)

Devotees are the primary users of the system who use the platform to browse temples and book darshan tickets. They interact mainly with the booking features and personal dashboard.

Responsibilities

Registration and Authentication

- Devotees can create an account by providing basic details such as name, email, and password.
- They can securely log in to the platform to access booking features.
- Authentication ensures that only registered users can book darshan slots.

Profile Management

- Devotees can view and update their personal profile information.
- They can modify details such as name, email address, and password.

Temple Browsing

- Devotees can browse a list of available temples.
- They can view temple information such as location, description, and images.

Darshan Slot Booking

- Devotees can view available darshan slots for a selected temple.
- They can select their preferred date and time slot.
- They can book darshan tickets and receive a unique booking ID.

Booking Management

- Devotees can access a personal dashboard to view past and upcoming bookings.
- They can cancel bookings if cancellation is permitted by the temple.

Feedback and Experience Sharing

- Devotees can provide feedback and ratings about their darshan experience to help improve the service.

Logout

- Devotees can securely log out of the system after completing their activities.

---

2. Organizer (Temple Organizer)

Temple organizers manage temple-specific activities and ensure that darshan slots are properly scheduled and available for devotees.

Responsibilities

Organizer Profile Management

- Organizers can manage their profile details, including updating name, email, and password.

Temple Management

- Organizers can add and maintain temple information including temple name, location, description, and images.

Darshan Slot Management

- Organizers are responsible for creating and scheduling darshan slots for each temple.
- They can update slot timings, capacity, and availability.
- Special slots or priority slots can also be managed by organizers.

Booking Monitoring

- Organizers can view all bookings associated with their temples.
- They can manage bookings and adjust slot allocations if necessary.

Availability Management

- Organizers ensure that slot availability is updated correctly when bookings are made or canceled.

Notifications

- Organizers receive notifications related to booking activities and slot updates to manage temple operations efficiently.

Logout

- Organizers can securely log out of the system after completing management tasks.

---

3. Administrator (Admin)

Administrators have the highest level of access and are responsible for managing the overall system operations and maintaining platform integrity.

Responsibilities

System Management

- Admins oversee the entire temple darshan booking system.
- They ensure the proper functioning, security, and configuration of the platform.

Devotee Management

- Admins can create, update, and delete devotee accounts when required.
- They can monitor user activity and manage feedback or ratings provided by devotees.

Temple Organizer Management

- Admins manage temple organizer accounts.
- They can create, update, or remove organizer profiles and assign permissions.

Temple and Facility Management

- Admins can manage temple information and ensure correct data is maintained in the system.
- They can schedule or supervise darshan slot creation and resource allocation.

Event Management

- Admins can create and manage temple-related events.
- They can schedule or remove events depending on temple activities.

Reporting and Analytics

- Admins can generate reports related to bookings, temple popularity, and devotee activity.
- These reports help in analyzing system usage and improving temple services.

Logout

- Admins can securely log out of the system after completing administrative tasks.

USER FLOW

The flowchart illustrates the complete workflow of the temple darshan booking system, starting from a visitor accessing the platform to completing a booking or performing administrative actions.

---

1. Start

This is the entry point of the system workflow. It represents the moment when a user accesses the platform through a web browser or mobile application.

At this stage, the system initializes and prepares the landing page interface where users can explore available temple services.

---

2. Landing Page (Browse Temples)

The landing page is the main interface presented to visitors when they first enter the system.

Functions of this component:

- Displays a list of temples available in the system
- Shows basic information such as temple name, location, and images
- Allows users to browse temples without logging in
- Provides navigation options like login, registration, and temple search

This step allows even guest users to explore temple information before deciding to create an account.

---

3. View Temple Details

When a user selects a temple from the landing page, they are directed to the temple details page.

Information displayed includes:

- Temple name
- Temple location
- Description and historical details
- Temple images
- Available darshan slot timings
- Capacity and availability of slots

This page helps users decide whether they want to book a darshan slot for that temple.

---

4. Login / Register Decision

At this stage, the system checks whether the user is authenticated.

Two possibilities exist:

Guest User (No)

If the user is not logged in, the system will prompt them to log in or create an account before making a booking.

Since booking requires authentication, guest users cannot proceed further without registering.

Registered User (Yes)

If the user chooses to log in or register, they will proceed to the authentication process.

---

5. Login / Register

This component manages user authentication and account creation.

Login Functionality:

- Existing users enter their email and password
- The system verifies credentials using the database
- If valid, the user is granted access

Registration Functionality:

- New users provide details such as name, email, and password
- The system stores these details securely in the database
- After successful registration, users can log in

This step ensures secure user identification before accessing booking services.

---

6. User Role Identification

After successful login, the system determines the role assigned to the user.

Two roles exist in the system:

- Regular User (Devotee)
- Administrator

The system directs the user to the appropriate dashboard or functionality based on their role.

---

7. Regular User Path

If the user is a devotee, they can perform temple booking activities.

7.1 Select Temple

The user selects a temple from the available list.

Features:

- Browse temples
- Filter or search temples
- View temple availability

Once the temple is selected, the user proceeds to choose a darshan slot.

7.2 Choose Slot

The system displays available time slots for the selected temple.

Slot information includes:

- Date of darshan
- Time slot
- Available capacity

Users select a slot based on their preferred timing.

The system ensures that the slot still has available capacity before confirming the selection.

7.3 Confirm Booking

After selecting the slot, the user confirms their booking.

Actions performed:

- Booking record is created in the database
- Slot capacity is reduced accordingly
- A unique booking ID is generated
- Confirmation message is shown to the user

This step finalizes the darshan reservation.

7.4 View Bookings

Users can access their booking history through the booking dashboard.

Features:

- View upcoming bookings

- View past bookings
- Access booking details
- Track booking status

This helps users manage their reservations effectively.

7.5 Cancel Booking

If required, users can cancel an existing booking. **System actions during cancellation:**

- Booking status is updated to "Canceled"
- The slot capacity is increased again
- The booking is removed from active reservations

After cancellation, the user receives confirmation.

7.6 Logout

Users can securely log out from the system once their actions are complete.

Logging out ensures session security and prevents unauthorized access.

---

8. Admin Path

If the logged-in user is an administrator, they gain access to administrative functions.

8.1 Admin Dashboard

The admin dashboard is the control panel for system management.

Functions include:

- Managing temple information
- Managing slot availability
- Monitoring bookings
- Performing administrative updates

This dashboard allows admins to oversee the entire system.

8.2 Add / Edit / Delete Temple

Administrators manage temple information through this module.

Admin capabilities include:

- Adding new temples to the system
- Updating temple details
- Uploading temple images
- Removing temples if necessary

This ensures that temple data remains accurate and updated.

8.3 Manage Slots

Admins manage the darshan slot scheduling.

Slot management includes:

- Creating new slots
- Updating slot timings
- Setting capacity limits
- Deleting slots if needed

Proper slot management helps regulate visitor flow and avoid overcrowding.

8.4 Logout / End

After completing administrative tasks, the admin logs out of the system.

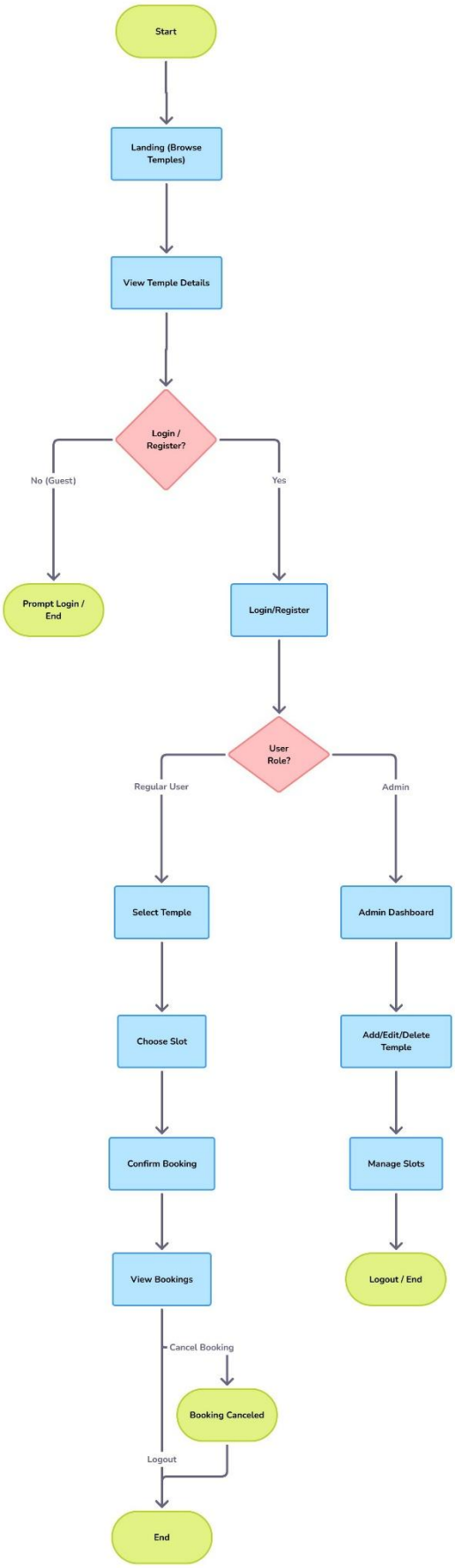
This ensures system security and proper session termination.

---

9. End

This represents the termination of the system workflow.

The session ends after the user or admin logs out or completes their actions within the system.


Start
Landing (Browse Temples)
View Temple Details
Login / Register?
No (Guest)
Yes
Prompt Login / End
Login/Register
User Role?
Regular User
Admin
Select Temple
Admin Dashboard
Choose Slot
Add/Edit/Delete Temple
Confirm Booking
Manage Slots
View Bookings
Logout / End
Cancel Booking
Booking Canceled
Logout
End

MVC Architecture

The Darshan Ease backend application follows the Model–View–Controller (MVC) architectural pattern, which separates the system into three main layers: Model, View, and Controller. This design pattern helps organize the codebase so that data handling, business logic, and request handling remain independent from each other.

Using MVC in the Darshan Ease application improves maintainability, scalability, and development efficiency, especially since the system manages user registration, profile management, and darshan slot bookings.

---

1. Model Layer (Data Layer)

The Model layer is responsible for managing the application data and database interactions. It defines how the data is structured and stored in the database.

In the Darshan Ease backend, models are implemented using Mongoose, an Object Data Modeling (ODM) library used with MongoDB.

Responsibilities of the Model Layer

- Defines data schemas for different entities in the system.
- Performs CRUD operations (Create, Read, Update, Delete).
- Ensures data validation and integrity.
- Communicates directly with the database.

Examples of Models in Darshan Ease

- User Model
    - Stores user information such as name, email, password, and profile details.
- Darshan Slot Model
    - Contains information about available darshan timings, slot capacity, and availability.
- Booking Model
    - Stores details of booked darshan tickets, user ID, slot ID, and booking status.

These models act as the data structure foundation of the entire application.

---

2. Controller Layer

The Controller layer contains the business logic of the application. It acts as an intermediary between the routing layer (View) and the Model layer.

When a request is received from the client, the controller:

1. Processes the request data.
2. Performs validation if required.
3. Calls the appropriate model functions.
4. Sends the response back to the client.

Responsibilities of Controllers

- Handle incoming API requests.
- Validate user input.
- Implement application logic.
- Interact with models to retrieve or store data.
- Return structured responses (JSON).

Examples of Controllers in Darshan Ease

- Authentication Controller
    - Handles user registration and login.
- Profile Controller

    - Manages profile updates and user details.
- Booking Controller
    - Processes darshan slot bookings and cancellations.

For example:

- When a user books a darshan slot, the controller checks slot availability, records the booking in the database, and sends confirmation.

---

3. View Layer (Routing Layer)

In a REST API backend, the View layer is represented by routes or API endpoints. These routes define how the backend responds to different HTTP requests.

Routes are implemented using Express.js, a web framework for Node.js.

Responsibilities of the View (Routing) Layer

- Define API endpoints.
- Map HTTP requests to controller functions.
- Structure the communication between client and server.

Example Routes in Darshan Ease

HTTP Method	Endpoint	Function
POST	/register	Register a new user
POST	/login	User authentication
GET	/slots	View available darshan slots
POST	/book	Book a darshan slot
PUT	/profile	Update user profile

For example:

- When a user sends a POST request to /book, the route forwards the request to the Booking Controller, which processes the booking using the Booking Model.

Advantages of Using MVC in Darshan Ease

1. Separation of Concerns

Each layer has a specific responsibility:

- Models manage data.
- Controllers handle business logic.
- Routes manage API endpoints.

This makes the code cleaner and easier to maintain.

2. Scalability

New features such as temple donation systems, VIP darshan, or notifications can be easily added by creating new models, controllers, and routes.

3. Reusability

Controllers and models can be reused across multiple endpoints or services, reducing redundant code.

4. Easier Testing

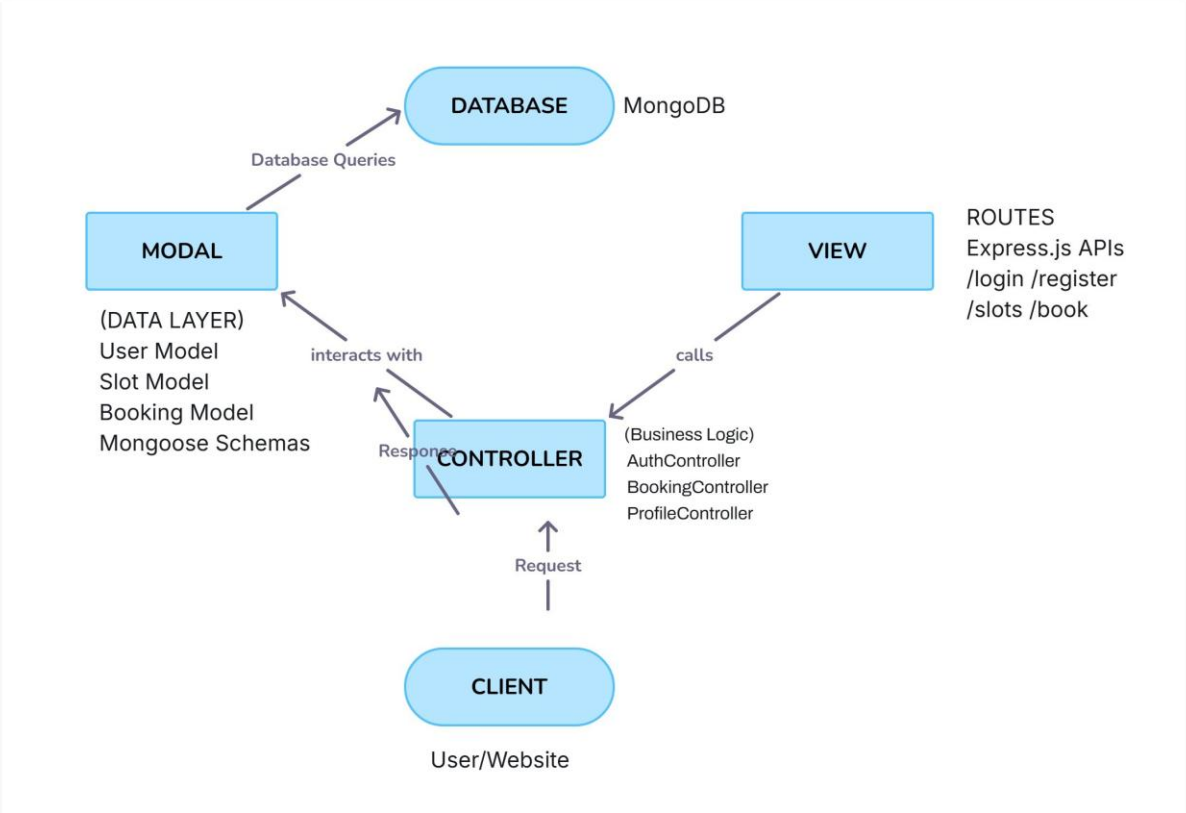
Developers can independently test:

- database models
- controller logic
- API endpoints

This improves reliability and debugging.

5. Collaboration-Friendly Development

Since the architecture is modular:

- Backend developers can work on controllers.
- Database engineers can work on models.
- API developers can work on routes.

This enables efficient teamwork during development.